SETS, FUNCTIONS, DOMAIN & RANGE

- $(a,b]$: a excluded, b included. $A\cap B$ = in both; $B\setminus A$ = in B but not A (count elements!). $\{1,2,3\}\cap(1,4]=\{2,3\}$.
- $(f\circ g)(x)=f(g(x))$ (do inside first). $g\circ f$ needs range$(f)\subseteq$dom(g). On a graph: find $f(2)$ first, then feed it back in. Open dot = not in function.
- **Domain rules:** denominator $\neq 0$; $\sqrt{}\geq 0$; $\sqrt{}$ in denominator >0; $\ln(\cdot)>0$; $\tan x$: $x\neq\frac{\pi}{2}+k\pi$; x^x, ln: need $x>0$; $\sin\frac{1}{1+x^2}$ fine $\forall x$.
- Range of $a\cos(bx+c)+d$ (also sin) $=[d-|a|,d+|a|]$. x^2+k on $[1,\infty)$: start at value at $x=1$.
- 2 vars: domain is a region in the xy-plane. "defined at $xy=0$": no $\ln(xy)$, $1/\sqrt{xy}$, $1/x$, $1/y$, ($\frac{xy}{x+y}$ ok at xy=0 unless x+y=0).

CONTINUITY

- Continuous at a: $\lim_{x\to a^-}f=\lim_{x\to a^+}f=f(a)$. Piecewise: plug the boundary into BOTH pieces, set equal, solve for k (quadratic in $k\Rightarrow$ may be 2 values; check all boundaries, solutions must satisfy all).
- Differentiable $\Rightarrow$ continuous (not converse). **Not differentiable at:** corner/kink ($|x-a|$), cusp ($x^{2/3}$, $\sqrt{|x|}$), vertical tangent ($x^{1/3}$), jump/hole, vertical asymptote (tan at $\frac{\pi}{2}$, $\frac{1}{x+2}$ at -2). $(x-1)^{3/2}$ is differentiable at 1; $(x+3)^{1/3}$ NOT at -3.

EXPONENTS & LOGS

- $a^m a^n=a^{m+n}$, $(a^m)^n=a^{mn}$, $a^{-n}=1/a^n$, $a^{1/n}=\sqrt[n]{a}$, $(\frac{a}{b})^{-1}=\frac{b}{a}$.
- $\log_a(xy)=\log_a x+\log_a y$; $\log_a\frac{x}{y}=\log_a x-\log_a y$; $\log_a x^n=n\log_a x$. $\log_a a=1$, $\log_a 1=0$, $a^{\log_a x}=x$, $\ln e^x=x$.
- **Change of base:** $\log_a x=\frac{\ln x}{\ln a}$. $\log_{a^n}x=\frac{1}{n}\log_a x$ ($\log_{16}=\frac{1}{2}\log_4$, $\log_{64}=\frac{1}{3}\log_4$, $\log_9=\frac{1}{2}\log_3$, $\log_4=\frac{1}{2}\log_2$). Convert everything to the target base, then combine powers.
- $\ln(x^2)=2\ln|x|$. Logs of sums don't split.

LIMITS & ASYMPTOTES

- Plug in. If $\frac{0}{0}$: factor & cancel / multiply by conjugate. $\frac{k}{0}$: $\pm\infty$ (VA).
- **Standard:** $\frac{\sin x}{x}\to 1$, $\frac{\sin ax}{ax}\to 1$, $\frac{1-\cos x}{x}\to 0$, $\frac{1-\cos x}{x^2}\to\frac{1}{2}$, $\frac{e^x-1}{x}\to 1$, $\frac{\ln(1+x)}{x}\to 1$, $\frac{\tan x}{x}\to 1$. e.g. $\frac{\sin x(1-\cos x)}{x^2}\to 1\cdot 0=0$.
- $\lim_{x\to\infty}$ rational: compare degrees. deg num < deg den: $y=0$; equal: ratio of leading coeffs; num bigger: no HA. $e^{-x}\to 0$, $\frac{\ln x}{x}\to 0$.
- **Vertical asymptote:** den $=0$ AFTER cancelling common factors. A cancelled factor is a *hole*, not an asymptote. Count VA + HA (HA at $\pm\infty$ may be same line, count once). e.g. $\frac{x+2}{(x+1)(x+2)}$: VA $x=-1$, HA y=0 $\Rightarrow$ 2. $\frac{x^2+x-6}{x^2-x-2}=\frac{(x+3)(x-2)}{(x-2)(x+1)}$: VA $x=-1$, HA y=1 $\Rightarrow$ 2. $\frac{x^2-9}{x-3}=x+3$ for $x\neq 3$: hole at 3, no VA.

TRIG (RADIANS) — LEARN THIS

π rad $=180°$; deg$\to$rad: $\times\frac{\pi}{180}$. $\sin^2+\cos^2=1$; $1+\tan^2=\sec^2$; $\tan=\frac{\sin}{\cos}$.

Trick: sin of $0,\frac{\pi}{6},\frac{\pi}{4},\frac{\pi}{3},\frac{\pi}{2}$ = $\frac{\sqrt{0}}{2},\frac{\sqrt{1}}{2},\frac{\sqrt{2}}{2},\frac{\sqrt{3}}{2},\frac{\sqrt{4}}{2}$; cos = same list reversed. tan = sin / cos.

Quadrants (ASTC): I all +; II only sin +; III only tan +; IV only cos +. Reduce to reference angle in QI, then fix the sign.

°	rad	sin	cos	tan	°	rad	sin	cos	tan
0	0	0	1	0	180	π	0	-1	0
30	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	210	$\frac{7\pi}{6}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45	$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	225	$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$	1
60	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	240	$\frac{4\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$\sqrt{3}$
90	$\frac{\pi}{2}$	1	0	und.	270	$\frac{3\pi}{2}$	-1	0	und.
120	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$	300	$\frac{5\pi}{3}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$-\sqrt{3}$
135	$\frac{3\pi}{4}$	$\frac{1}{\sqrt{2}}$	$-\frac{1}{\sqrt{2}}$	-1	315	$\frac{7\pi}{4}$	$-\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	-1
150	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{3}}$	330	$\frac{11\pi}{6}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{\sqrt{3}}$

$\sin(-x)=-\sin x$, $\cos(-x)=\cos x$; $\sin(\pi-x)=\sin x$, $\cos(\pi-x)=-\cos x$; $\sin(x+\pi)=-\sin x$, $\cos(x+\pi)=-\cos x$; $\sin(\frac{\pi}{2}-x)=\cos x$, $\cos(\frac{\pi}{2}-x)=\sin x$. Period 2π (tan: π).

$\sin 2x=2\sin x\cos x$; $\cos 2x=\cos^2x-\sin^2x=1-2\sin^2x=2\cos^2x-1$.

$\sin^2x=\frac{1-\cos 2x}{2}$, $\cos^2x=\frac{1+\cos 2x}{2}$.

$\sin(a\pm b)=\sin a\cos b\pm\cos a\sin b$; $\cos(a\pm b)=\cos a\cos b\mp\sin a\sin b$.

$\sec=\frac{1}{\cos}$, $\operatorname{cosec}=\frac{1}{\sin}$, $\cot=\frac{1}{\tan}$.

arcsin, arctan outputs: $[-\frac{\pi}{2},\frac{\pi}{2}]$; arccos: $[0,\pi]$.

VECTORS

- $\mathbf{u}\cdot\mathbf{v}=u_1v_1+u_2v_2+u_3v_3=|\mathbf{u}||\mathbf{v}|\cos\theta$ (scalar). $|\mathbf{u}|=\sqrt{\mathbf{u}\cdot\mathbf{u}}$. $\cos\theta=\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$. Perpendicular $\iff\mathbf{u}\cdot\mathbf{v}=0$. Acute angle $\iff$ dot >0 (graph: arrows with angle $>90°$ give negative dot).
- Unit vector $\hat{\mathbf{u}}=\mathbf{u}/|\mathbf{u}|$; opposite direction: $-\mathbf{u}/|\mathbf{u}|$. Check: $(2,1,-1)$, $|\cdot|=\sqrt{6}$.
- $\text{proj}_{\mathbf{v}}\mathbf{u}=\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v}$. If $|\mathbf{v}|=1$, $\mathbf{u}\cdot\mathbf{v}=|\mathbf{u}|\cos\theta$: test options!
- **Cross** $\mathbf{u}\times\mathbf{v}=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\u_1&u_2&u_3\\v_1&v_2&v_3\end{vmatrix}=(u_2v_3-u_3v_2,\ u_3v_1-u_1v_3,\ u_1v_2-u_2v_1)$ (middle term sign flipped!). $\perp$ to both; $|\mathbf{u}\times\mathbf{v}|=|\mathbf{u}||\mathbf{v}|\sin\theta$; $\mathbf{u}\times\mathbf{v}=-\mathbf{v}\times\mathbf{u}$; $\mathbf{u}\times\mathbf{u}=\mathbf{0}$; parallel $\iff\mathbf{u}\times\mathbf{v}=\mathbf{0}$.
- **Triangle area** $=\frac{1}{2}|\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}|$; parallelogram $=|\mathbf{u}\times\mathbf{v}|$. Triple product $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$ = parallelepiped volume (= 0 if coplanar).
- **Defined?** scalar·vector is NOT a dot or cross: $(\mathbf{u}\cdot\mathbf{v})\times\mathbf{w}$ undefined; $(\mathbf{u}\cdot\mathbf{v})\mathbf{w}$ fine (scalar mult.); $\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w})$ fine; $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}$ fine; $(\mathbf{u}\times\mathbf{u})\cdot(\mathbf{v}\times\mathbf{v})=0$ fine; $\mathbf{u}\cdot\mathbf{v}\cdot\mathbf{w}$ undefined.
- **Line:** $\mathbf{r}=\mathbf{p}+t\mathbf{d}$. Through A,B: $\mathbf{d}=B-A$, any point on line as $\mathbf{p}$, any multiple of $\mathbf{d}$ ok (check point lies on it!). Parallel lines: $\mathbf{d}$'s multiples. Perpendicular lines: $\mathbf{d}_1\cdot\mathbf{d}_2=0$.
- **Plane:** $ax+by+cz=d$, normal $\mathbf{n}=(a,b,c)$; $\mathbf{n}\cdot(\mathbf{r}-\mathbf{r}_0)=0$. From 3 pts: $\mathbf{n}=\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}$. Parallel planes: $\mathbf{n}$ multiples. Line $\parallel$ plane $\iff\mathbf{d}\cdot\mathbf{n}=0$. Line $\perp$ plane $\iff\mathbf{d}\parallel\mathbf{n}$. Line of intersection of 2 planes: $\mathbf{d}=\mathbf{n}_1\times\mathbf{n}_2$ + a common point.
- **Line ∩ plane:** sub x,y,z of line into plane eq, solve t: one $t\to$ a point (plug back); 0 = nonzero $\to$ none; 0 = 0 $\to$ whole line. Dist. point$\to$plane $=\frac{|ax_0+by_0+cz_0-d|}{|\mathbf{n}|}$.

MATRICES

- $(m\times n)(n\times p)=m\times p$ (inner must match). $(AB)^T=B^TA^T$; $(AB)^{-1}=B^{-1}A^{-1}$; $(ABC)^{-1}=C^{-1}B^{-1}A^{-1}$; $(A^T)^T=A$. A^TB with both 4×2: $2\times4\cdot4\times2=2\times2$. A^TBB^T with A,B 4×2: $(2\times4)(4\times2)(2\times4)=2\times4$. $AB\,B^TA^T$ (3×2,2×3): 3×3.
- **2×2 inverse** of $\begin{bmatrix}a&b\\c&d\end{bmatrix}$: swap $a\leftrightarrow d$, negate b,c, divide by $ad-bc$: $\frac{1}{ad-bc}\begin{bmatrix}d&-b\\-c&a\end{bmatrix}$. Check answers: $AA^{-1}=I$ (test 1 row/col).
- **Determinant:** 2×2 $ad-bc$. 3×3: expand along row/col with most zeros, signs checkerboard $+-+/-+-/+-+$; triangular $\to$ product of diagonal. Zero row/col, or row multiple of other $\to\det=0$.
- $\det(kA)=k^n\det A$ ($n\times n$; 3×3: k^3). $\det(AB)=\det A\det B$; $\det A^T=\det A$; $\det A^{-1}=\frac{1}{\det A}$; $\det A^k=(\det A)^k$. Row swap: det $\times(-1)$; row$\times c$: det $\times c$; R_i+cR_j: no change. E.g. $\det(2A^2B)=2^3\cdot2^2\cdot3=96$ for $\det A$=2, $\det B$=3 (3×3). $|2AB^T|=8\cdot3\cdot2=48$.
- **Invertible** $\iff\det\neq0\iff$ RREF $=I\iff$ rank $n\iff Ax=0$ only $x=0\iff Ax=b$ unique soln. Any zero row $\Rightarrow$ NOT invertible.
- **Valid row ops:** swap $R_i\leftrightarrow R_j$; $R_i\to cR_i$ ($c\neq0$); $R_i\to R_i+cR_j$ ($j\neq i$). Invalid: new R_i not containing old R_i (e.g. $R_3=R_2-2R_1$), or scaling by 0.

LINEAR SYSTEMS

- Reduce $[A|b]$. Leading 1s = pivots; columns without pivot = free variables (parameters s,t). **Unique soln:** pivot in every variable column. **Infinite:** consistent & ≥1 free variable (solution = point + $s\mathbf{d}$, line; two params = plane). **None:** row $[0\ 0\ 0\,|\,c]$, $c\neq0$.
- With parameter k: find values making a pivot vanish, plug that k in, reduce, inspect rows. Row $[0\ 0\ k^2-1\,|\,k-1]$: k=1 $\to$ 0=0 (good); $k=-1\to0=-2$ (none). Row $[0\ k-1\ 0\,|\,2]$ at k=1: 0=2 none. Do each row!
- If A^{-1} exists, $Ax=b\Rightarrow x=A^{-1}b$ (exactly 1 solution). $\det A=0$: 0 or ∞ solutions. Homogeneous $Ax=0$ always consistent (trivial soln).
- RREF with a free variable: read each pivot row as "var = const − (free-var terms)"; set free var = s; write answer as $\mathbf{p}+s\mathbf{d}$. Check with a point.
- **3 planes (in $\mathbb{R}^3$):** unique pt $\to$ "pin-

wheel" meeting at one point (rank 3); common line → "book" (rank 2, consistent); no soln → 3 parallel sheets, or triangular prism, or 2 parallel + 1 cutting (rank < aug. rank); all same plane → rank 1. Upper-triangular (staircase) system with nonzero diagonal → one point.

DIFFERENTIATION

- **First principles:** $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ (put $x+h$ in EVERY x: $xe^x \to \frac{(x+h)e^{x+h}-xe^x}{h}$). Given limit form $\frac{1}{h}[\frac{1}{(x+h)^2}-\frac{1}{x^2}] \Rightarrow g=\frac{1}{x^2}$. $\frac{(x-1)-(x+h-1)}{(x-1)(x+h-1)} = \frac{1}{x+h-1}-\frac{1}{x-1} \Rightarrow f = \frac{1}{x-1}$.
- $(x^n)' = nx^{n-1}$, $(e^x)' = e^x$, $(a^x)' = a^x \ln a$, $(\ln x)' = \frac{1}{x}$, $(\log_a x)' = \frac{1}{x\ln a}$, $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$, $(\sec x)' = \sec x\tan x$, $(\operatorname{cosec} x)' = -\operatorname{cosec} x\cot x$, $(\cot x)' = -\operatorname{cosec}^2 x$, $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$, $(\arctan x)' = \frac{1}{1+x^2}$.
- Product $(fg)' = f'g+fg'$; Quotient $(\frac{f}{g})' = \frac{f'g-fg'}{g^2}$; **Chain** $(f(g))' = f'(g)\cdot g'$: $(\cos e^{3x})' = -3e^{3x}\sin(e^{3x})$; $(e^{2x})' = 2e^{2x}$; $(e^x\sin(3x^2-1))' = e^x(\sin(3x^2-1)+6x\cos(3x^2-1))$. $h\cdot\frac{f}{g}$: $h'\frac{f}{g}+h\frac{f'g-fg'}{g^2}$.
- **Implicit:** differentiate both sides wrt x; $\frac{d}{dx}y^n = ny^{n-1}y'$, $\frac{d}{dx}(xy) = y+xy'$, $\frac{d}{dx}f(y) = f'(y)y'$. Collect y' terms, solve. $x^3+2xy+y^2=4 \Rightarrow y' = -\frac{3x^2+2y}{2x+2y}$. $x-\cos y = 5 \Rightarrow 1+\sin y\, y' = 0$.
- **Tangent:** $y-y_0 = m(x-x_0)$, $m = f'(x_0)$ (point of tangency must satisfy curve $y_0 = f(x_0)$; for implicit, find y' at the pt). Normal slope $-1/m$.

APPLICATIONS OF f', f''

- **Critical point:** $f'=0$ or f' undefined, but x must be in domain of f. Count zeros of numerator + zeros of denominator (where f defined); repeated factor = one point. $x^{3/5}$ in denominator: x=0 critical (if $f(0)$ exists).
- $f'>0$: increasing; $f'<0$: decreasing. Test sign at one point per interval (test point must be INSIDE the interval).
- **1st-deriv. test:** f' + → −: local max; − → +: local min; no sign change: neither (e.g. $(x-1)^2$ factor, no sign change).
- **2nd-deriv. test:** $f'(c)=0$: $f''>0$ min; $f''<0$ max; $f''=0$ inconclusive. $f''>0$: concave up (smile); $f''<0$: concave down. f' increasing $\iff f''>0$.
- **Inflection:** concavity CHANGES (sign of f'' differs either side; $f''=0$ alone is not enough; f defined there). Test: $f''(0)$ and $f''(2)$ opposite signs for pt at 1.
- Graph clues: sharp V point = continuous, non-diff, local min (like $|x|$); f smooth peak: max. $f = \frac{x}{1+x^2}$: $f' = \frac{1-x^2}{(1+x^2)^2}$, zeros $\pm1 \Rightarrow$ 2 extrema. $f = x^5+x^4$: $f'' = 20x^3+12x^2 = 4x^2(5x+3)$, $f''<0$ on $x<-\frac{3}{5}$ only. Cubic $-2x^3+9x^2-12x+5$: $f' = -6(x-1)(x-2)>0$ on $(1,2)$. $-2x^3+6$ at 0: $f'=0$, no sign change → stationary inflection.
- Local max/min on graph of f': where f' crosses zero: + → − = max of f, − → + = min. Graph of f: slope sign gives f'; bending gives f'' (peak shape: $f''<0$).

INTEGRATION

- $\int x^n = \frac{x^{n+1}}{n+1}$ $(n\neq -1)$; $\int\frac{1}{x} = \ln|x|$; $\int e^{ax} = \frac{e^{ax}}{a}$; $\int a^x = \frac{a^x}{\ln a}$; $\int\sin ax = -\frac{\cos ax}{a}$; $\int\cos ax = \frac{\sin ax}{a}$; $\int\sec^2 x = \tan x$; $\int\sec x\tan x = \sec x$; $\int\tan x = -\ln|\cos x|$; $\int\frac{1}{1+x^2} = \arctan x$; $\int\frac{1}{\sqrt{1-x^2}} = \arcsin x$; $\int\ln x = x\ln x - x$; $\int\frac{1}{ax+b} = \frac{1}{a}\ln|ax+b|$. Always $+C$.
- $\int\frac{f'}{f} = \ln|f|$. $\frac{2x}{x^2-1} \to \ln|x^2-1|+C$; with $f(2)=0$: $\ln(x^2-1)-\ln 3$.
- **u-sub:** choose u = inside function whose derivative appears; $du = u'dx$. Definite: change limits (x=0 → u=$u(0)$), e.g. $\int_0^{\pi/4} 2\cos 2x\sin^2 2x\,dx$, $u=\sin 2x$: $\int_0^1 u^2 du$. $\int\frac{e^x}{\sqrt{2-e^x}}$: $u = 2-e^x$ gives $-2\sqrt{2-e^x}$. $\int(8x-12)(4x^2-12x)^4 = \frac{(4x^2-12x)^5}{5}$.
- **By parts:** $\int u\,dv = uv - \int v\,du$. Choose u in order **LIATE** (Log, Inverse trig, Algebraic, Trig, Exp). $\int x^n\sin/\cos/e^{ax}$: needs n rounds ($x^3\sin 5x$: 3 times). $\int x^n\ln x = \frac{x^{n+1}}{n+1}\ln x - \int\frac{x^n}{n+1}$. $\int xe^{ax} = e^{ax}(\frac{x}{a}-\frac{1}{a^2})$. $\int x\cos(ax+b) = \frac{x}{a}\sin(\cdot)+\frac{1}{a^2}\cos(\cdot)$. $\int x\sin(ax+b) = -\frac{x}{a}\cos(\cdot)+\frac{1}{a^2}\sin(\cdot)$. Check by differentiating!
- **Definite:** $\int_a^b f = F(b)-F(a)$ (constants cancel). $\int_a^b f = -\int_b^a f$; $\int_a^c = \int_a^b+\int_b^c$; $\int(f+g) = \int f+\int g$; $\int cf = c\int f$. $\int_{-a}^a$ odd = 0; even = $2\int_0^a$. $\int_{-2}^2\sin x\,dx = 0$.
- **Area** between curve & x-axis: find roots, split, take $|\cdot|$ of each part (area above minus area below counts positive). Between curves: $\int$(top − bottom). From a graph: count triangles/rectangles (below axis negative for $\int$).
- $\int_0^1(6x^2+\sin x) = 2+1-\cos 1 = 3-\cos 1$. $\int_{\pi/6}^{\pi/3}(1-2\cos x) = [x-2\sin x] = \frac{\pi}{6}+1-\sqrt{3}$. Linear combos: $4(3)+a(-2)-6(4)=0 \Rightarrow a=-6$.

FUNCTIONS OF 2 VARIABLES

- **Partials:** f_x: treat y as constant (and vice versa). $f = xe^{x+y}$: $f_x = (1+x)e^{x+y}$, $f_y = xe^{x+y}$. $h = (y^2+1)e^x$: $h_x+h_y = e^x(y^2+2y+1)$. $\ln(x^2y^2) = 2\ln|x|+2\ln|y|$ (simplify first!): $f = (2x+7)\ln(x^2y^2) \Rightarrow f_y = \frac{2(2x+7)}{y}$.
- $f_x = f_y$: set expressions equal, test options. $z = x^2+2y^2-xy$: $2x-y = 4y-x \iff 3x=5y \Rightarrow (5,3)$.
- **Tangent plane at (a,b):** $z = f(a,b)+f_x(a,b)(x-a)+f_y(a,b)(y-b)$. $z = \ln(x^2+y^2)$ at $(0,1)$: $f=0, f_x=0, f_y=2 \Rightarrow z = 2y-2$. $z = 2x^2y$ at $(1,2)$: $f=4, f_x=4xy=8, f_y=2x^2=2 \Rightarrow z = 4+8(x-1)+2(y-2) = 8x+2y-8$.
- **Level curve:** set $f(x,y)=c$ and solve for y. $3x^2+y=c$: parabola $y=c-3x^2$ (opens down). x^2+y^2: circles. y^2+2x^2: ellipses (tall if x has larger coeff.). $ax+by$: lines. xy: hyperbolas. $y+2x^2$: parabolas.
- Domain: $(\ln(-x-y))^2$ needs $-x-y>0 \iff y<-x$ (below line $y=-x$): region A (in 3rd quad) yes, B (above axis) no.

EXAM TACTICS

Test the options (plug a number) when algebra is slow; check dimensions first; $\sqrt{}$ & ln domain traps; sign errors in cross product; − sign in chain rule for cos; hole vs asymptote; k^n det; all values of k.